

UPPSALA UNIVERSITET

- Decision trees
- Ensemble methods
  - Bagging
  - Random forests
  - (Gradient) Boosting

Machine learning – Block 2

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This week's lecture

- Trees
- Bagging
- Random Forest
- Boosting (Trees)



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Practicalities

- Assignment submission out of sync
  - Keep it? Or move to wednesdays 23.59?



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Section 1

Decision trees



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Decision trees: basic idea

- Have you ever played the game "20 questions"?



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Decision trees: basic idea

- Have you ever played the game "20 questions"?
- Decision trees is more or less that game!



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Decision trees: basic idea

- Have you ever played the game "20 questions"?
- Decision trees is more or less that game!
- In the case of classification (prediction), the idea is to classify the new observation by
  1. Asking a questions
  2. Based on the previous answer, ask new question
  3. Questions are asked until a conclusion is reached



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Decision trees: basic idea

New observation: a Komodo dragon. What class does the tree predict?

Name	Body temp	Live birth	Legs	Class
Human	warm-blooded	yes	yes	mammal
Whale	warm-blooded	yes	no	mammal
Cat	warm-blooded	yes	yes	mammal
Cow	warm-blooded	yes	yes	mammal
Python	cold-blooded	no	no	reptile
Turtle	cold-blooded	no	yes	reptile
Salmon	cold-blooded	no	no	fish
Eel	cold-blooded	no	no	fish
Pigeon	warm-blooded	no	yes	bird
Penguin	warm-blooded	no	yes	bird

1. Does it give live birth?



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1. Does it give live birth? (No!)
2. Is it warm-blooded?



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1. Does it give live birth? (No!)
2. Is it warm-blooded? (No!)
3. Does it have legs?



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1. Does it give live birth? (No!)
2. Is it warm-blooded? (No!)
3. Does it have legs? (Yes!)

Predicted class: reptile.

Regression trees: The regions of a tree

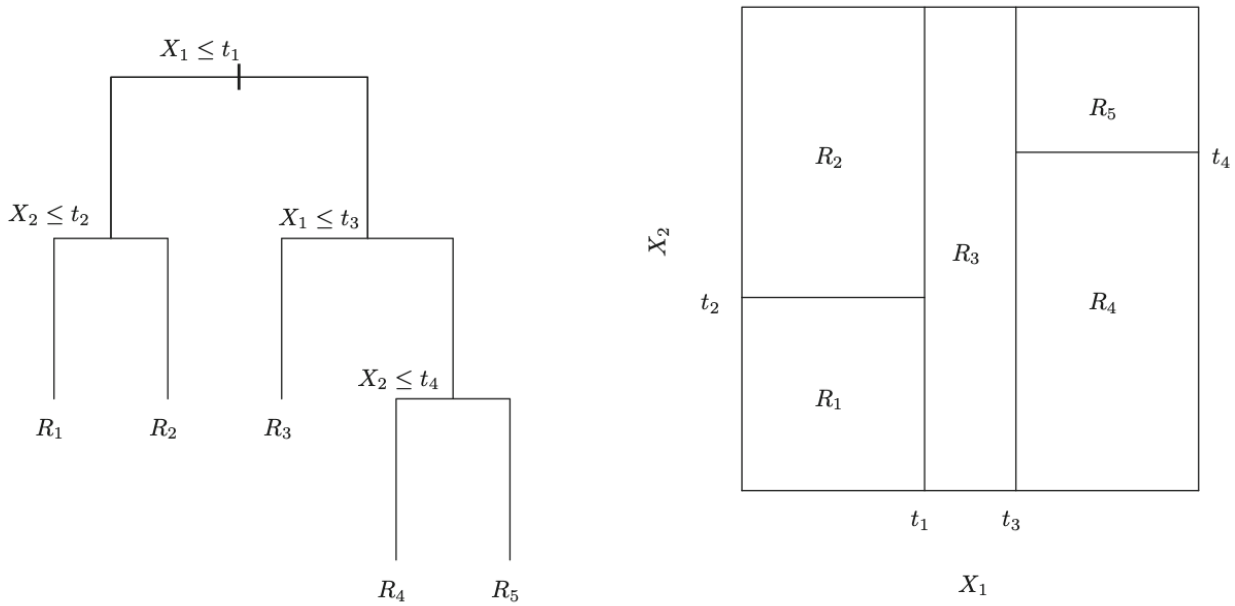


Figure: Regions of a tree (James et al., 2013, Fig. 8.3)



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Regression Trees

$$T(x) = \sum_{m=1}^{M} \gamma_m I(x \in R_m),$$

where M is the total number of regions and $I(x \in R_m)$ is an indicator variable if x_i belongs to region R_m and γ_m is the prediction for region m.



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Regression Trees

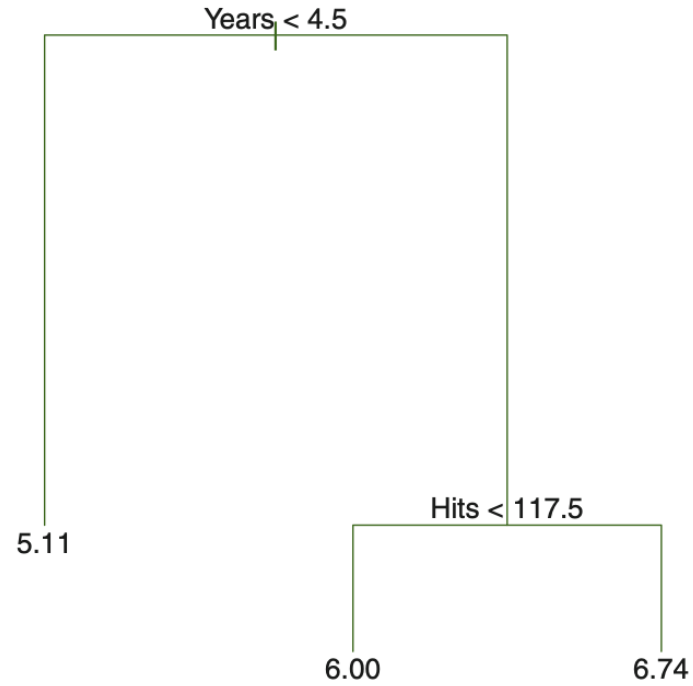


Figure: Regression Tree (James et al., 2013, Fig. 8.1)

- The textbook tree predicts log salary. Assignment 2 uses `Salary` directly, so its leaf values are on a different scale.
- Each leaf displays its predicted log salary: the mean log salary of the training observations in that leaf.

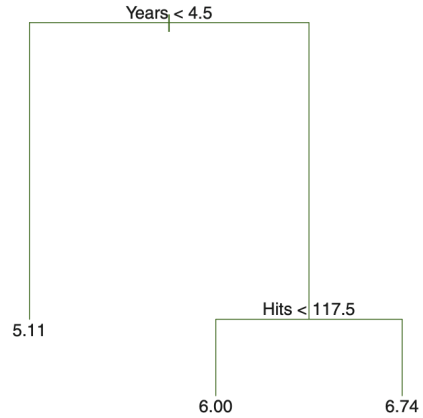


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Pair discussion: read the Hitters tree

A baseball player has six years in the league and 100 hits.



James et al. (2013), Fig. 8.1. Go left if the condition is true.

Pairs, 1–2 minutes:

1. Which branches does the player follow? What is the prediction?
2. What does the number in that leaf represent?

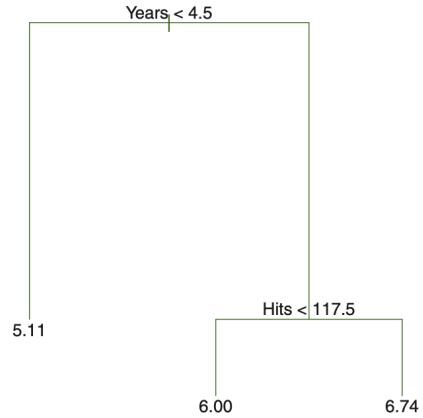


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2. What does the number in that leaf represent?

Answer

Right at `Years`, then left at `Hits`: 6.00, the mean log salary of the training players in that leaf.



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Regression Trees

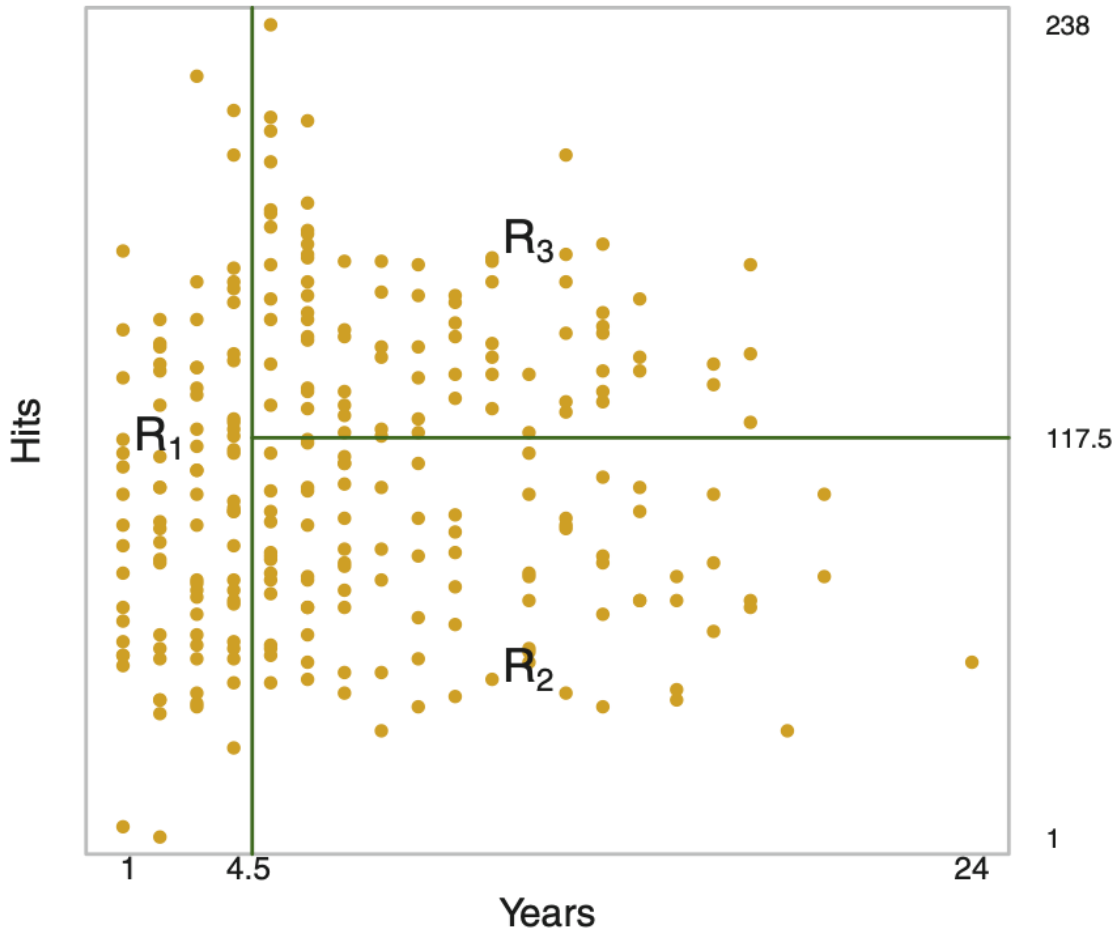


Figure: `Hitters` data and regression tree regions (James et al., 2013, Fig. 8.2)



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Estimating a Decision Tree

1. A tree has two groups of parameters $\Theta = (\gamma, R)$ that we need to learn.



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Estimating a Decision Tree

1. A tree has two groups of parameters $\Theta = (\gamma, R)$ that we need to learn.
2. We want a tree that minimize $L(\theta) = \sum_i^N (y_i - T_\Theta(x_i))^2$, here the squared loss



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2. We want a tree that minimize $L(\theta) = \sum_i^N (y_i - T_\Theta(x_i))^2$, here the squared loss
3. Usually we estimate γ_m as the mean of y_i in the region as:

$$\hat{\gamma}_m = \frac{1}{N_m} \sum_{x_i \in R_m}^{N_m} y_i \, ,$$

where N_m is the number of observations in region R_m.



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$$\hat{\gamma}_m = \frac{1}{N_m} \sum_{x_i \in R_m}^{N_m} y_i \, ,$$

   where N_m is the number of observations in region R_m.
4. Learning R_m exact is generally computationally infeasable so we use a greedy heuristic.



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How to do a split?

Let S be the set of training row indices in the current node. For predictor j and threshold s, form

$$R_1(j,s) = \{i \in S : x_{ij} \le s\}, \qquad R_2(j,s) = \{i \in S : x_{ij} > s\}.$$

A valid split has $|R_1| \ge l$ and $|R_2| \ge l$. For each valid split, predict the child means $\hat{c}_m = |R_m|^{-1} \sum_{i \in R_m} y_i$.

Choose (j,s) to minimise $$\sum_{i \in R_1} (y_i - \hat{c}_1)^2 + \sum_{i \in R_2} (y_i - \hat{c}_2)^2.$$

Assignment 2 tests observed values. A midpoint between consecutive values yields the same training split as the lower value, but can route new values differently.

Adapted from ESL, Eqs. (9.12–9.14), with the minimum child size used in Assignment 2.



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Worked example: scoring a split

One predictor, six observations in the current node, and minimum child size $l = 2$:

x_i	1	2	3	4	5	6
y_i	1	3	2	7	9	8

For $x \leq 3$ versus $x > 3$, the child means are 2 and 8.

$$\text{Left SSE} = (1-2)^2 + (3-2)^2 + (2-2)^2 = 2,$$

$$\text{Right SSE} = (7-8)^2 + (9-8)^2 + (8-8)^2 = 2.$$



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Threshold s	Child sizes	Total child SSE
1	$(1, 5)$	Invalid
2	$(2, 4)$	31
3	$(3, 3)$	4
4	$(4, 2)$	21.25
5	$(5, 1)$	Invalid
6	$(6, 0)$	Invalid

Skip any split with fewer than two rows in either child. Choose $s = 3$: its total SSE of $2 + 2 = 4$ is the lowest among valid splits.



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Pair discussion: score a candidate split

Consider this node with one predictor:

x_i	1	2	3	4
y_i	2	4	8	9

Compare $s = 1.5$ and $s = 2.5$. Send $x_i \leq s$ left and $x_i > s$ right. Predict each child's mean and add the squared errors from both children.

Pairs, 1–2 minutes:

1. Which threshold has lower total child SSE?
2. Which of these thresholds is valid if each child needs at least two observations?



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Pairs, 1–2 minutes:

1. Which threshold has lower total child SSE?
2. Which of these thresholds is valid if each child needs at least two observations?

Answer

$s = 1.5$: child means 2 and 7; SSE $= 0 + 9 + 1 + 4 = 14$. Child sizes $(1, 3)$.
$s = 2.5$: child means 3 and 8.5; SSE $= 1 + 1 + 0.25 + 0.25 = 2.5$. Child sizes $(2, 2)$.
Choose $s = 2.5$: it has lower SSE and is the only valid split.



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How to do a split? Pseudo-code

Input: $\mathbf{X}$ with N_s rows and P predictors, $\mathbf{y}$, and minimum child size l.

1. Initialise SS and S as $N_s \times P$ matrices filled with ∞.
2. For each predictor j and candidate $s = x_{k,j}$:
   - 2.1 Form $R_1 = \{i : x_{i,j} \leq s\}$ and $R_2 = \{i : x_{i,j} > s\}$.
   - 2.2 If $|R_1| < l$ or $|R_2| < l$, skip this candidate; leave its $SS_{k,j} = \infty$.
   - 2.3 Compute the response means $\hat{c}_1$ and $\hat{c}_2$ in the two children.
   - 2.4 Store $S_{k,j} = s$ and

$$SS_{k,j} = \sum_{i \in R_1} (y_i - \hat{c}_1)^2 + \sum_{i \in R_2} (y_i - \hat{c}_2)^2.$$

3. If no finite score exists, return `NULL`.
4. Otherwise, choose any $(k_{\text{final}}, j_{\text{final}})$ attaining the minimum SS; tied minima are acceptable.
5. Return the chosen j, s, and child row indices R_1, R_2 (and optionally SS).



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Growing a Decision Tree: Greedy Algorithm

Each queue entry holds original data rows r, a parent tree row, and the pointer to fill (`R1_i` or `R2_i`).

1. Call `tree_split(X, y, l)` at the root. If it returns `NULL`, return one leaf with prediction mean(y). Otherwise, write tree row 1 with both pointers set to -1, then queue its left and right children.
2. Repeat steps 2–4 until the queue is empty: remove the first entry and call
   `split = tree_split(X[r, , drop = FALSE], y[r], l)`.
3. Append a new tree row k. Store the split variable and threshold, or, for `NULL`, store a leaf with prediction mean($y[r]$). Set the parent pointer from the queue entry to k in either case.
4. If the node splits, set its two child pointers to -1 (not yet written). Append `r[split$R1]` and `r[split$R2]` to the back of the queue, with parent k and their respective pointer names.

Example ($l = 1$): For data rows $(4, 9, 15)$, local split indices $(1, 3)$ and (2) map to $(4, 15)$ and (9). If this is the root's left child and is written as tree row 2, set `R1_i[1] = 2`. The root's pending right child stays ahead of both new entries.



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Decision trees: Classification Trees

- How do we do if we have a classification tree?



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Decision trees: Classification Trees

- How do we do if we have a classification tree?
- Let $p(j \mid t)$ be the training proportion of class j in node t.
- Gini impurity for J classes is

$$G(t) = 1 - \sum_{j=1}^{J} p(j \mid t)^2.$$

It is zero for a pure node and largest when the proportions are equal.



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  It is zero for a pure node and largest when the proportions are equal.

- Choose the valid split that minimises the size-weighted child impurity:

$$\frac{n_L}{n} G(t_L) + \frac{n_R}{n} G(t_R), \qquad n = n_L + n_R.$$



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- At a leaf, report the class proportions or predict the most frequent class.



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CART

CART stands for Classification and Regression Trees.

- Grow a binary tree by greedily choosing splits: reduce SSE for regression or impurity for classification.



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- Grow a binary tree by greedily choosing splits: reduce SSE for regression or impurity for classification.
- Use stopping rules to limit growth, for example by requiring a minimum child size.



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CART

CART stands for Classification and Regression Trees.

- Grow a binary tree by greedily choosing splits: reduce SSE for regression or impurity for classification.
- Use stopping rules to limit growth, for example by requiring a minimum child size.
- In the grow-then-prune approach, start with a large tree and use cost-complexity pruning to select a smaller subtree.

Assignment 2 implements greedy splitting and a minimum child size; it does not require post-pruning.



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Controlling tree size: regularisation

- Depth: the maximum number of splits on a path from the root to a leaf.
- Leaf size: the number of training observations in a leaf.

Deep trees with small leaves can fit noise and vary substantially across training datasets.



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Deep trees with small leaves can fit noise and vary substantially across training datasets.

- Stop during growth: limit depth or require a minimum child size. In Assignment 2, a split is allowed only if both children contain at least l observations.
- Prune after growth: grow a large tree, then remove branches. CART's cost-complexity penalty charges for each leaf.



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- Prune after growth: grow a large tree, then remove branches. CART's cost-complexity penalty charges for each leaf.

As in L1b, use validation performance to choose how much complexity to allow.



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Cost-Complexity Pruning

Goal: Reduce overfitting by shrinking a large tree into a simpler subtree.

For any subtree T, define:

- Training error:

$$R(T) = \sum_{m=1}^{|T|} \sum_{i \in R_m} (y_i - \hat{\gamma}_m)^2$$

- Tree size: $|T|$ = number of terminal/leaf nodes/regions
- Full loss:

$$R_\alpha(T) = R(T) + \alpha|T|$$

$\alpha \geq 0$ controls the penalty:

- $\alpha = 0$: minimises training error alone, risking overfitting.
- Large α: favours smaller trees, risking underfitting.



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Weakest-Link Pruning Algorithm

1. Grow a large tree T_0.
2. For each internal node t, compare its current subtree T_t with replacing it by a single leaf:

$$g(t) = \frac{R(t) - R(T_t)}{|T_t| - 1}.$$

   Here $R(t)$ is the training SSE after replacement. The ratio is the extra SSE per leaf removed.
3. At $\alpha = \min_t g(t)$, prune the branches attaining this minimum.
4. Recompute the ratios and repeat to get nested subtrees:

$$T_0 \supset T_1 \supset T_2 \supset \cdots \supset T_K$$

5. Grow and prune on each CV training fold; compare α values on the held-out folds. Use the selected α to prune the tree grown on all training data.



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Section 2

Ensemble methods



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Wisdom of the crowd

Simulated example (Prize academy, see ESL):

1. 50 members vote in 10 categories, each with 4 nominations.



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2. For any category, only 15 voters have some knowledge ($p > 0.25$)



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1. 50 members vote in 10 categories, each with 4 nominations.
2. For any category, only 15 voters have some knowledge ($p > 0.25$)
3. For each category, the 15 experts are chosen at random from the 50.

'Individual' = a single weak learner/member

'Consensus' = the nominee with the most votes



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Wisdom of the crowd

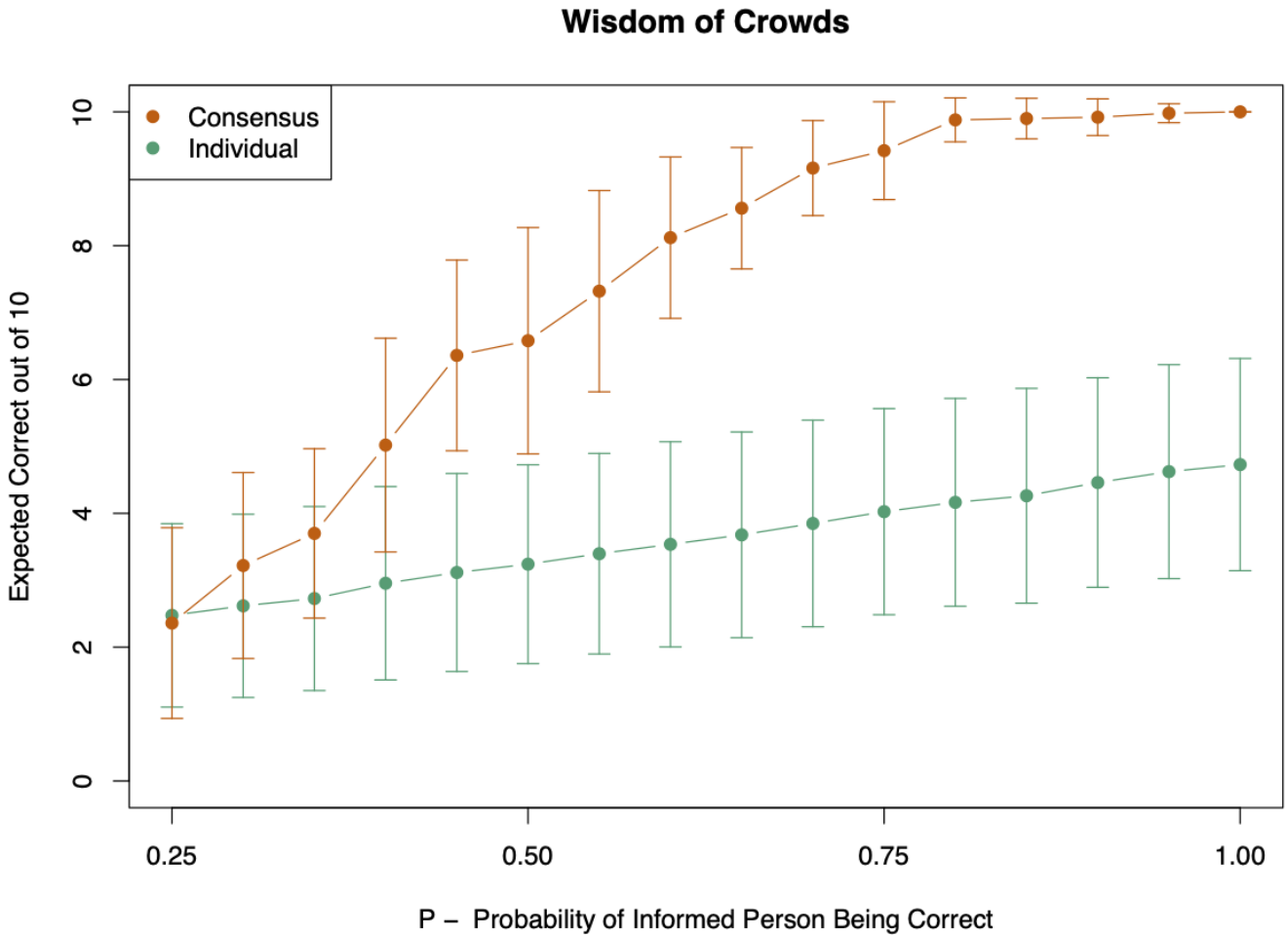


Figure: Simulated Award Voting, Fig. 8.11 (ESL)



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General idea of ensembles

The idea of an ensemble is simple: If it difficult to find one really good model perhaps we can find several weaker models and combine their predictions.



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General idea of ensembles

The idea of an ensemble is simple: If it difficult to find one really good model perhaps we can find several weaker models and combine their predictions.

A simple example: Say you have one outcome Y and 4 covariates X_1, X_2, X_3, X_4. The goal is to predict Y. A possible ensemble would be to fit

$$
\begin{aligned}
y &= \alpha_1 + \beta_1 X_1 + \epsilon_1 \\
y &= \alpha_2 + \beta_2 X_2 + \epsilon_2 \\
y &= \alpha_3 + \beta_3 X_3 + \epsilon_3 \\
y &= \alpha_4 + \beta_4 X_4 + \epsilon_4
\end{aligned}
$$

and then use the mean of their predictions

$$
\hat{y}_{\text{ensemble}} = \frac{1}{4}\sum_{b=1}^{4} \hat{y}_b = \frac{1}{4}\sum_{b=1}^{4} (\hat{\alpha}_b + \hat{\beta}_b X_b). \tag{1}
$$



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Two key parts of an ensemble

1. The prediction models (sometimes called 'learners')
   - A single model in an ensemble can be a simple or a complex model
   - Often the ensemble contains many simple models.



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Two key parts of an ensemble

1. The prediction models (sometimes called 'learners')
   - A single model in an ensemble can be a simple or a complex model
   - Often the ensemble contains many simple models.
2. The weighting of each prediction in the final ensemble prediction
   - Many different algorithms for weighting together the predictions from many models
   - Models/learners with better predictive power can be given larger weights in the final prediction



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Ensambles of decision trees

- A common type of ensembles is ensembles of decision trees.
- We will focus on this case, but note that any type of model can be included in an ensemble in principle.



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Bagging and Boosting

Recall the decomposition for expected squared prediction error:

$$\text{Expected squared error} = \text{bias}^2 + \text{variance} + \text{irreducible error}.$$



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Bagging and Boosting

Recall the decomposition for expected squared prediction error:

$$\text{Expected squared error} = \text{bias}^2 + \text{variance} + \text{irreducible error}.$$

- A deep tree can fit complex patterns, but may change substantially when the training data change.
- A shallow tree may be more stable, but can miss important patterns.



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Bagging and Boosting

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- A deep tree can fit complex patterns, but may change substantially when the training data change.
- A shallow tree may be more stable, but can miss important patterns.

Bagging: Averaging unstable trees often reduces variance.

Boosting: Adding corrections from shallow trees can reduce bias.

Both methods can affect bias and variance.



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Subsection 1

Bagging



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Bagging (Bootstrap AGGregating)

- Train several deep trees and combine their results
- Use bootstrap to train different trees

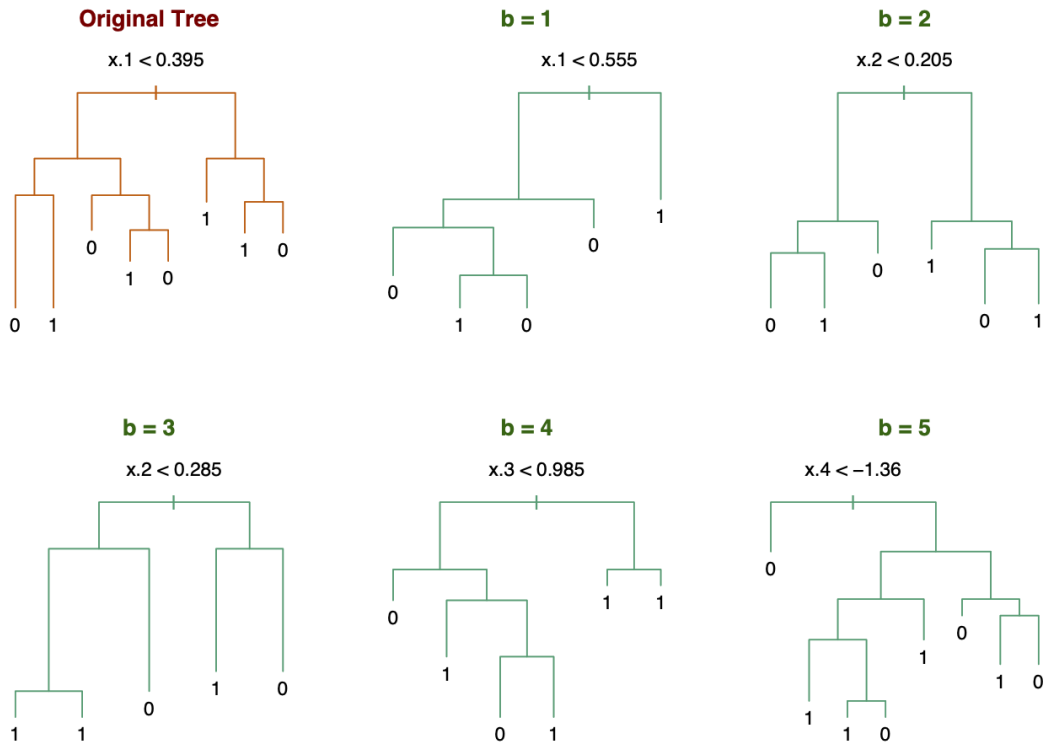


Figure: Bagging trees, Fig. 8.9 (ESL)



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Bagging (Bootstrap AGGregating)

1. Draw, *with replacement*, a random sample of N units from the original sample
2. Fit a prediction model (e.g., a deep decision tree)
3. Repeat steps 1-2 B times
4. Weight together the predictions from the B models into a final ensemble prediction as

$$\hat{f}_{\text{bag}}(x_i) = \frac{1}{B}\sum_b \hat{f}^b(x_i) = \frac{1}{B}\sum_b \hat{T}(x_i|\Theta_b)$$



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Pair discussion: why randomise the predictors?

Suppose you fit many trees to different bootstrap samples. One predictor is so strong that almost every tree uses it at the root.

Pairs, 1–2 minutes:

1. Why might averaging many more trees help less than you expect?
2. What might happen if that predictor is sometimes unavailable when choosing a split? Think about both the individual trees and the ensemble.



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Pairs, 1–2 minutes:

1. Why might averaging many more trees help less than you expect?
2. What might happen if that predictor is sometimes unavailable when choosing a split? Think about both the individual trees and the ensemble.

Answer

Trees that make similar errors leave less room for averaging to help. Sampling predictors can make trees more diverse, so their errors may cancel more. It can also weaken individual splits by hiding a strong predictor.



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Subsection 2

Random forests



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Random Forest

- A bagging ensemble method, but...



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Random Forest

- A bagging ensemble method, but...
- Each split considers m of the p available predictors.



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Random Forest

- A bagging ensemble method, but...
- Each split considers m of the p available predictors.
- Common starting values are $m = \lfloor \sqrt{p} \rfloor$ for classification and $m = \max(1, \lfloor p/3 \rfloor)$ for regression. Treat m as a tuning parameter.



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Random Forest: Algorithm

1. For each of B trees, draw N training observations with replacement.
2. Grow the tree on that bootstrap sample. At **each node, including the root**, draw a fresh subset of m predictors without replacement.
3. Search for the best valid split among those predictors. If there is no valid split, make a leaf.
4. For regression, average the B tree predictions:

$$\hat{f}_{\text{rf}}(x) = \frac{1}{B} \sum_{b=1}^{B} \hat{T}(x \mid \Theta_b).$$

With $m = p$, this reduces to bagging under the same tree-growing rules.



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Random forest variance: an illustration

At a fixed input x, suppose each tree prediction has variance σ^2 and each pair has correlation ρ, across training datasets and tree draws.

$$\mathbb{V}\big(\hat{f}_{\mathrm{rf}}(x)\big) = \sigma^2 \left[\rho + \frac{1-\rho}{B}\right].$$



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$$\mathbb{V}\big(\hat{f}_{\mathrm{rf}}(x)\big) = \sigma^2 \left[\rho + \frac{1-\rho}{B}\right].$$

- If $\rho = 0$, averaging B trees reduces the variance to σ^2/B.
- If $\rho > 0$, variance approaches $\rho\sigma^2$ as B grows.



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Why can tree predictions be correlated?

- Given a fixed training dataset, independently randomised trees give independent predictions at a fixed x.
- Across repeated training datasets, predictions can move together: all trees in a forest are built from the same underlying dataset.



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Bagging vs. Random forest

- When a predictor is particularly strong, many bagged trees may choose it near the root.



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Bagging vs. Random forest

- When a predictor is particularly strong, many bagged trees may choose it near the root.
- Sampling candidate predictors at each node can make forest trees less similar and reduce correlation.



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- Decision trees
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Bagging vs. Random forest

- When a predictor is particularly strong, many bagged trees may choose it near the root.
- Sampling candidate predictors at each node can make forest trees less similar and reduce correlation.
- Smaller m can improve diversity, but may hide useful predictors at a split and make individual trees less accurate.
- Choose m based on predictive performance, not correlation alone.



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Subsection 3

(Gradient) Boosting



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Boosting

1. In boosting, models/trees are trained sequentially



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Boosting

1. In boosting, models/trees are trained sequentially
2. Each new model tries to target weak spots of the previous models



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Boosting (binary AdaBoost)

Use labels and base predictions $y_i, h_b(x_i) \in \{-1, +1\}$. Start with score $F_0(x) = 0$ and weights $w_i = 1/N$.

1. At round b, fit h_b using the weights and compute its weighted error:
$$e_b = \frac{\sum_i w_i I(y_i \neq h_b(x_i))}{\sum_i w_i}.$$
2. If $e_b = 0$, stop and use the perfect training learner (the infinite-weight limit). If $e_b \geq 1/2$, discard the learner and stop.
3. Otherwise, add the learner and upweight misclassified observations:
$$\alpha_b = \log \frac{1 - e_b}{e_b},$$
$$F_b(x) = F_{b-1}(x) + \alpha_b h_b(x),$$
$$w_i \leftarrow w_i \exp\big(\alpha_b I(y_i \neq h_b(x_i))\big).$$
4. Predict the sign of the final score; break zero-score ties in favour of $+1$.



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Boosting example

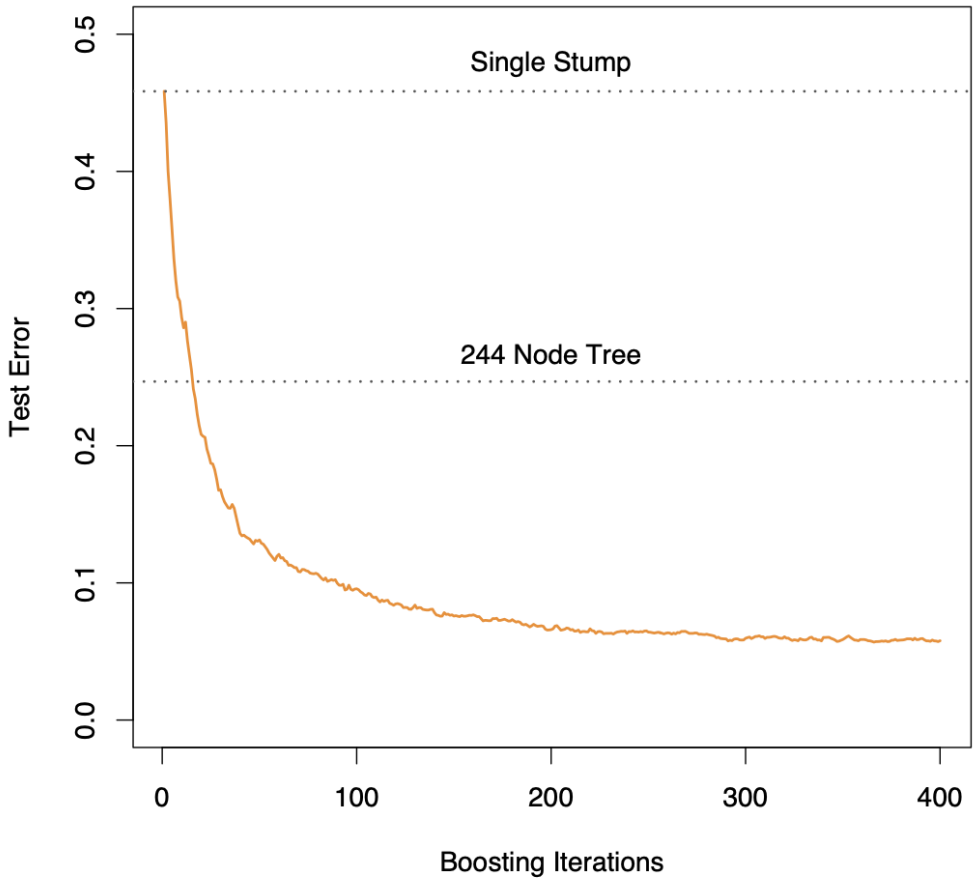


Figure: Boosting example (using stumps as h_b), Fig. 10.2 (ESL)



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Boosting trees: start with squared loss

For $L(y, f) = \frac{1}{2}(y - f)^2$, start at the mean response $f_0(x) = \bar{y}$.
At round b:

1. Compute residuals $r_{ib} = y_i - f_{b-1}(x_i)$.
2. Fit a shallow regression tree T_b to (x_i, r_{ib}).
3. Add a fraction of its prediction:

$$f_b(x) = f_{b-1}(x) + \nu T_b(x), \qquad 0 < \nu \leq 1.$$



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$$f_b(x) = f_{b-1}(x) + \nu T_b(x), \qquad 0 < \nu \leq 1.$$

Fitting to residuals targets the stagewise loss $\sum_{i=1}^{N} L\big(y_i, f_{b-1}(x_i) + T_b(x_i)\big)$ before applying the learning rate ν.



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Pair discussion: take one boosting step

Two observations have outcomes 8 and 3. The current model predicts 5 for both. Use squared loss.

Pairs, 1–2 minutes:

1. What residual targets should the next tree fit?
2. If the tree predicts those targets exactly and the learning rate is 0.5, what are the updated predictions?



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1. What residual targets should the next tree fit?
2. If the tree predicts those targets exactly and the learning rate is 0.5, what are the updated predictions?

Answer

The residual targets are $8 - 5 = 3$ and $3 - 5 = -2$. Add half of each correction: $5 + 0.5(3) = 6.5$ and $5 + 0.5(-2) = 4$.



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Gradient Boosting Trees: general loss

Start with the constant minimising the training loss. At each round:

1. Compute pseudo-responses from the negative loss gradient:

$$r_{ib} = -\left.\frac{\partial L(y_i, z)}{\partial z}\right|_{z=f_{b-1}(x_i)}.$$

   For half squared error these are the ordinary residuals.
2. Fit a regression tree T_b to (x_i, r_{ib}).
3. Choose a step for the original loss, for example by minimising

$$\sum_{i=1}^{N} L\big(y_i, f_{b-1}(x_i) + aT_b(x_i)\big)$$

   over a. Then set $f_b(x) = f_{b-1}(x) + \nu a T_b(x)$. Tree boosting can instead optimise a separate update in each leaf.



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Bagging, random forests, and boosting

Bagging Fit each tree separately on a bootstrap sample of rows. All predictors are eligible at each split. Average the tree predictions for regression.



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Random forests Fit trees on bootstrap samples too, but draw a fresh subset of candidate predictors at each node. Average the tree predictions for regression.

Boosting Fit trees sequentially. Each tree targets what the current ensemble gets wrong, and its prediction is added as a correction.



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Comparisons

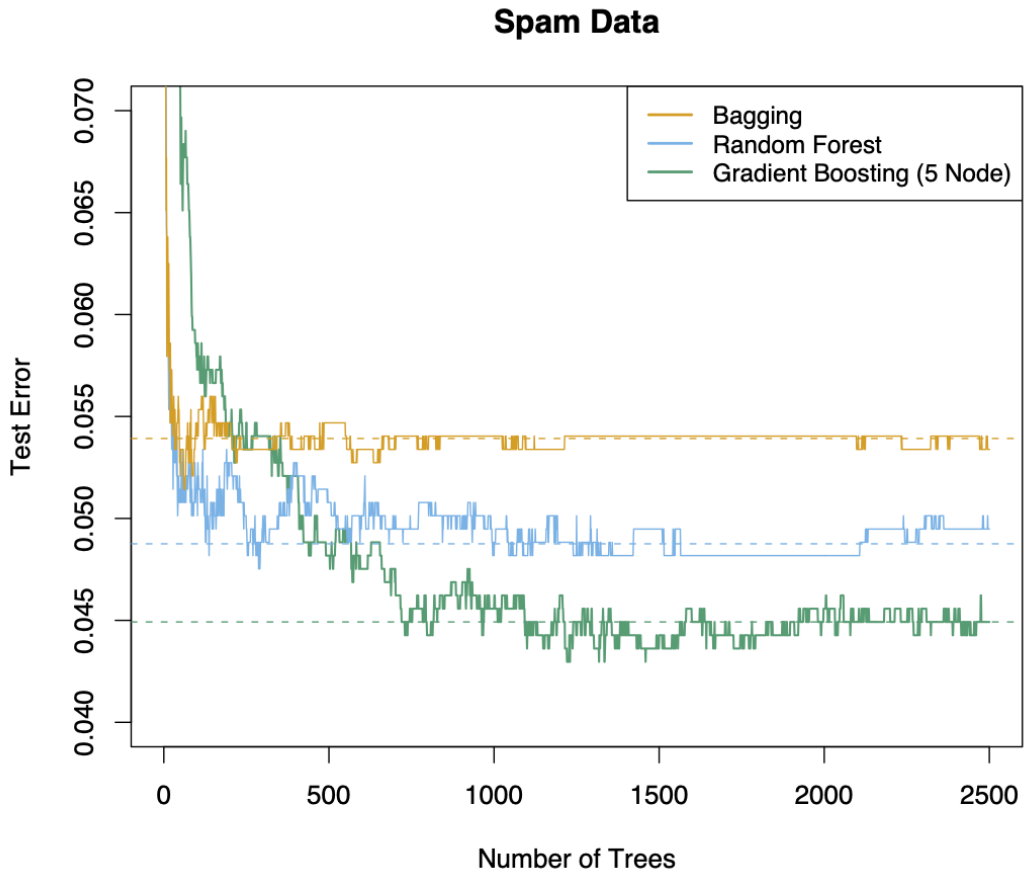


Figure: Test error on one spam dataset (ESL, Fig. 15.1). The ranking can differ on other datasets.



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XGBoost

XGBoost implements gradient tree boosting with a penalty for each tree:

$$\sum_{i=1}^{N} L\big(y_i, f_{\text{boost}}(x_i)\big) + \sum_{b=1}^{B} \Omega(T_b).$$



XGBoost

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$$\sum_{i=1}^{N} L\big(y_i, f_{\text{boost}}(x_i)\big) + \sum_{b=1}^{B} \Omega(T_b).$$

For example, use the penalty

$$\Omega(T_b) = \lambda_1 Q_b + \frac{\lambda_2}{2}\|\gamma_b\|_2^2, \qquad \lambda_1, \lambda_2 \geq 0.$$

Here Q_b is the number of leaves and γ_b is the vector of leaf scores in tree b.

- The first term penalises extra leaves.
- The second shrinks large leaf scores, as in ridge regularisation.

Chen and Guestrin (2016), Section 2.1.